Python Course

Introductory

Introduction
- What is a programming Language?
- Python Interpreter
- How to run python code
- Python 2 vs python3
- Why so many languages

Developer Environment
- Install python 3 on window and LINUX
- Python developer tools (Sublime text, Visual studio code, pycharm , Jupyter notebook)
- Optional terminal commands

Python Basics (1)
- Python Data Types
- Math functions
- Operator precedence
- Variables
- Expression vs statements
- Escape sequence
- Immutability

Python Basics (2)
- Breaking the flow
- Conditional logic
- Indentation in python
- Truthy vs Falsey
- Ternary operator
- Short circuiting
- For loops
- While loops
- Break, continue, pass
- Functions
- Parameters and arguments
- Default parameters and keywork arguments
- Return
- Docstring
- Clean code
- Args and kwargs
- Scope rules
- Global keyword, nonlocal keyword

Advance

Object Oriented programming
- What is OOP ?
- @classmethod and @staticmethod
- Encapsulation
- Abstraction
- Private vs public variables
- Inheritance
- Polymorphism
- Super
- Object introspection
- Dunder method
- Method resolution order (Order)

Error Handing
- How to debug code.
- Error in python
- Error Handing

Python Decorators
- Why do we need decorator?
- High order functions
- decorators

Python Generators
- Generators
- Generator performance
- Under the hood of generators

Module in python
- Modules in python
- Packages in python
- Python built in modules
- Python package index
- Pip install
- Virtual environments

Web Development with Python
- What is framework
- Talking about flask
- Talking about django

Machine Learning + Data Science
- What is machine learning and data science ?
- Python tools in machine learning
- Talking about Pandas, Numpy, Tensorflow